

Esercizio I: [4 pt]

Si consideri la funzione

$$f(x, y) = \begin{cases} \frac{x^{\frac{1}{3}} y^2}{x^2 + y^6} \sin(x) & \text{se } (x, y) \neq (0, 0) \\ 0 & \text{se } (x, y) = (0, 0) \end{cases}$$

Stabilire se la funzione è differenziabile in $(0, 0)$.

a) Continuità

$$y^3 = x$$

$$\lim_{y \rightarrow 0} \frac{y^{\cancel{3}}}{2y^{\cancel{6}}} = \frac{1}{2}$$

Non è diff perché non è continua

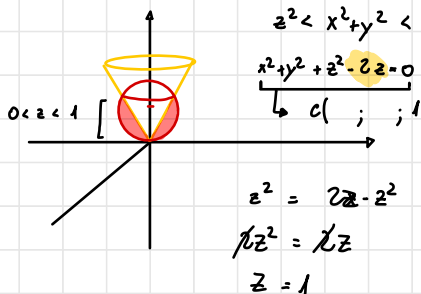
Esercizio II: (pt 8)

Calcolare

$$\iiint_A \frac{1}{\sqrt{x^2 + y^2 + z^2}} dx dy dz$$

dove $A = \{(x, y, z) \in \mathbb{R}^3, z^2 \leq x^2 + y^2 \leq 2z - z^2\}$.

$$\begin{cases} x = p \cos \theta \\ y = p \sin \theta \\ z = z \end{cases}$$



$$c = \left(-\frac{a}{2}\right)$$

$$c = 0, 0, 1$$

$$r = \sqrt{1^2} = 1$$

$$\iiint_A \frac{1}{\sqrt{p^2 + z^2}} p dp d\theta dz$$

$$\int_0^1 \int_0^{2\pi} \int_z^{\sqrt{2z-z^2}} \frac{1}{\sqrt{p^2 + z^2}} p dp d\theta dz$$

$$2p (p^2 + z^2)^{-\frac{1}{2}}$$

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$$(p^2 + z^2)^{\frac{1}{2}}$$

$$= \frac{1}{2} \int_0^1 \int_0^{2\pi} \frac{(p^2 + z^2)^{\frac{1}{2}}}{\frac{1}{2}} d\theta dz$$

$$= \int_0^1 \int_0^{2\pi} \left| (p^2 + z^2)^{\frac{1}{2}} \right|_{\frac{1}{2}}^{\sqrt{2z-z^2}} d\theta dz$$

$$= \int_0^1 \int_0^{2\pi} (2z)^{\frac{1}{2}} - (z^2)^{\frac{1}{2}} d\theta dz$$

$$\frac{1}{2}$$

$$= 2\pi \left[\int_0^1 (2z)^{1/2} dz - \int_0^1 \sqrt{2z^2} dz \right]$$

$$= 2\pi \left[2 \left| \frac{2}{3} z^{3/2} \right|_0^1 - \frac{\sqrt{2}}{2} |z^2|_0^1 \right]$$

$$= 2\pi \left[2 \left| \frac{2}{3} \right| - \frac{\sqrt{2}}{2} (1) \right]$$

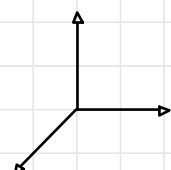
$$= 2\pi \left[\frac{4}{3} - \frac{\sqrt{2}}{2} \right]$$

$$= 2\pi \left[\frac{8 - 3\sqrt{2}}{6} \right]$$

$$= \frac{8 - 3\sqrt{2}}{3}$$

Esercizio III: (pt 6)

Calcolare l'area della superficie contenuta all'interno del cilindro di equazione $u^2 + v^2 = 1$ descritta parametricamente da $(2u, v, 1-2v)$.



$$r = (2u, v, 1-2v)$$

$$J = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix} = \begin{vmatrix} 2 & v & 0 \\ 0 & 1 & -2 \\ 0 & u & -2 \end{vmatrix} = -2v; +4, 2u$$

$$\|J\| = \sqrt{4v^2 + 16 + 4u^2} = 2\sqrt{v^2 + 4 + u^2}$$

$$\frac{\partial f}{\partial u} = 2, v, 0$$

$$\frac{\partial f}{\partial v} = 0, u, -2$$

$$Area = \iint_{u^2 + v^2 < 1} 2\sqrt{v^2 + u^2 + 4} \, du \, dv$$

$$= \iint \sqrt{p^2 + 4} \, p \, dp \, d\theta$$

$$= 2p(p^2 + 4)^{1/2}$$

$$2\pi \int_0^1 \left(\frac{2}{3} (p^2 + 4)^{3/2} \right)$$

$$= \frac{4\pi}{3} \left[5^{3/2} - 4^{3/2} \right]$$

$$V = -\log(x^2 + y^2)$$

$$V = -\log(x^2 + y^2)$$

$$L = \nabla V$$

$$= +\log(\quad)$$

$$= -\log(1) + \log(1)$$

$$= 0$$

$$r(t) = [\cos t, \sqrt{2} (\sin t)^2]$$

$$r_i = 1, 0 \quad \left(\frac{\sqrt{2}}{2}\right)^2$$

$$r_f = -\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} \quad \frac{2}{4}$$

$$\frac{1}{2}$$